

ACL Configuration Lab

1) **Title:** Access Control List (ACL) Configuration in Cisco Packet Tracer.

2) **Aim / Objectives :**

To configure standard ACL and restrict communication between different networks.

3) **Devices Used :**

- 2 Switch (2960)
- 1 Router (2911)
- 2 PCs
- Copper Straight-through Cables

4) **Topology :**

PC0 ---- Switch0 ---- Router0 ---- Switch1 ---- PC1

5) **IP Addressing Table:**

HR Network

Device	IP Address
PC0	192.168.10.2
Router G0/0	192.168.10.1

Finance Network

Device	IP Address
PC1	192.168.20.2
Router G0/1	192.168.20.1

6) Procedure / Steps :

Step 1 — Configure Router Interfaces

```
enable
configure terminal

interface g0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface g0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

Step 2 — Configure PCs

PC0

IP Address: 192.168.10.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

PC1

IP Address: 192.168.20.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.20.1

Step 3 — Test Before ACL

From PC0:

```
ping 192.168.20.2
```

Expected:

```
Reply from 192.168.20.2
```

Step 4 — Create ACL

Goal:

HR Network (192.168.10.0)

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Finance Network

Block

Router CLI:

```
access-list 10 deny 192.168.10.0 0.0.0.255
access-list 10 permit any
```

Step 5 — Apply ACL

```
Apply on G0/0:  
interface g0/0  
ip access-group 10 in  
exit
```

Step 6 — Verify

From PC0:

```
ping 192.168.20.2
```

Expected:

```
Request timed out
```

```
ACL working
```

Verification Commands

```
show access-lists
```

and

```
show running-config
```

7) Result :

Successfully configured a Standard ACL on the router to control network traffic and restrict access between networks based on security requirements.

8) Concepts Learned :

- Access Control Lists (ACL)
- Traffic Filtering
- Network Security
- Packet Filtering
- Access Restrictions
- Router Security

9) Cybersecurity Relevance :

ACLs are used to:

- Block Unauthorized Access
- Restrict Network Traffic
- Protect Sensitive System
- Implement Security Policies
- Control Communication Between Departments

10) Screenshot :

